

AP CALCULUS AB - UNIT 2

Guided Notes 2.1: Rates of Change and Tangent Lines



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we calculate average rate of change from formulas, tables, graphs, and contexts?

- Calculate average rate of change from formulas, tables, graphs, and contexts.
- Interpret secant-line slope and tangent-line slope with correct units.
- Use shrinking intervals to motivate and estimate instantaneous rate of change.

Key Knowledge, Vocabulary, and Notation

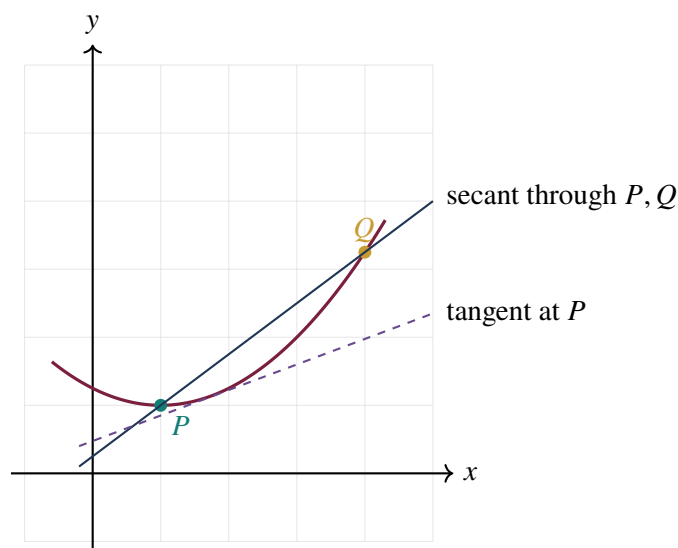
Average rate of change	$\frac{f(b) - f(a)}{b - a}$, the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.
Instantaneous rate of change	The limiting value of average rates as the interval collapses to a point; geometrically, the tangent-line slope.
Difference quotient	$\frac{f(a + h) - f(a)}{h}$ for $h \neq 0$.
Units	Output units divided by input units; for position in meters and time in seconds, the rate is meters per second.

Concept Development and Strategy

1. An average rate describes net change across an interval. It does not claim that the rate stayed constant inside the interval.
2. To approximate a rate at $x = a$, use secant slopes over intervals with endpoints increasingly close to a . A symmetric interval often reduces one-sided bias.
3. In contextual answers, state the quantity, sign, units, and time or input value. A negative rate means the dependent variable is decreasing with respect to the independent variable.

Shared Representation / Data

Rates of Change

**Worked Example: Secant slope from a formula**

For $s(t) = t^3 - 4t$ meters, find the average velocity on $[1, 3]$.

Answer and Reasoning

$s(3) = 15$ and $s(1) = -3$, so $\frac{15 - (-3)}{3 - 1} = 9$ meters per second.

Worked Example: A shrinking interval

For $f(x) = x^2 + 1$, calculate the average rate on $[2, 2 + h]$ and identify its limiting value as $h \rightarrow 0$.

Answer and Reasoning

$\frac{(2+h)^2 + 1 - 5}{h} = \frac{4h + h^2}{h} = 4 + h$. As $h \rightarrow 0$, the rate approaches 4.

Worked Example: Interpretation with units

The temperature $T(t)$ of a reaction is measured in $^{\circ}\text{C}$, where t is in minutes. Explain the meaning of $\frac{T(8) - T(5)}{3} = -1.6$.

Answer and Reasoning

From minute 5 to minute 8, the temperature decreased at an average rate of 1.6°C per minute.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Find the average rate of change of $f(x) = 3x^2 - 2$ on $[1, 4]$.

2. Find the average rate of change of $g(x) = \sqrt{x+1}$ on $[0, 8]$.

3. For $s(t) = t^3 - 6t^2 + 9t$, find the average velocity on $[1, 2]$ and on $[1, 1.5]$.

4. Use $f(x) = \frac{1}{x}$ to find the secant slope through $x = 2$ and $x = 2 + h$. Simplify.

5. For $f(x) = x^2 - 5x$, determine all intervals $[a, a + 2]$ on which the average rate is 3.

6. For $f(x) = \frac{x-1}{x+2}$, derive the average rate on $[a, a + h]$ in simplified form.

7. For $f(x) = |x|$, compute the average rates on $[-h, 0]$ and $[0, h]$ for $h > 0$. What do they predict at 0?

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. The average rate of change of f on $[a, b]$ is

(A) $\frac{f(a)-f(b)}{b-a}$

(B) $\frac{f(b)-f(a)}{b-a}$

(C) $f(b) - f(a)$

(D) $\frac{b-a}{f(b)-f(a)}$

2. For $f(x) = x^2$, the average rate on $[3, 3 + h]$ is

(A) 6

(B) $6 + h$

(C) $9 + h^2$

(D) $3 + h$

3. A calculator gives $f(1.999) = 6.994001$ and $f(2.001) = 7.006001$. A symmetric estimate of $f'(2)$ is

(A) 3

(B) 6

(C) 7

(D) 12

Common Errors and AP Precision

- State the derivative notation, input value, sign, and units when interpreting a rate in context.
- Check differentiability before using a tangent slope, and keep the order of Product and Quotient Rule terms accurate.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Suppose $\frac{f(a+h)-f(a)}{h} = 2a + 3 + h \sin(1/h)$ for $h \neq 0$. Determine the instantaneous rate at a .

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